

Computer Network and Communication CNS3200



Learning Outcome

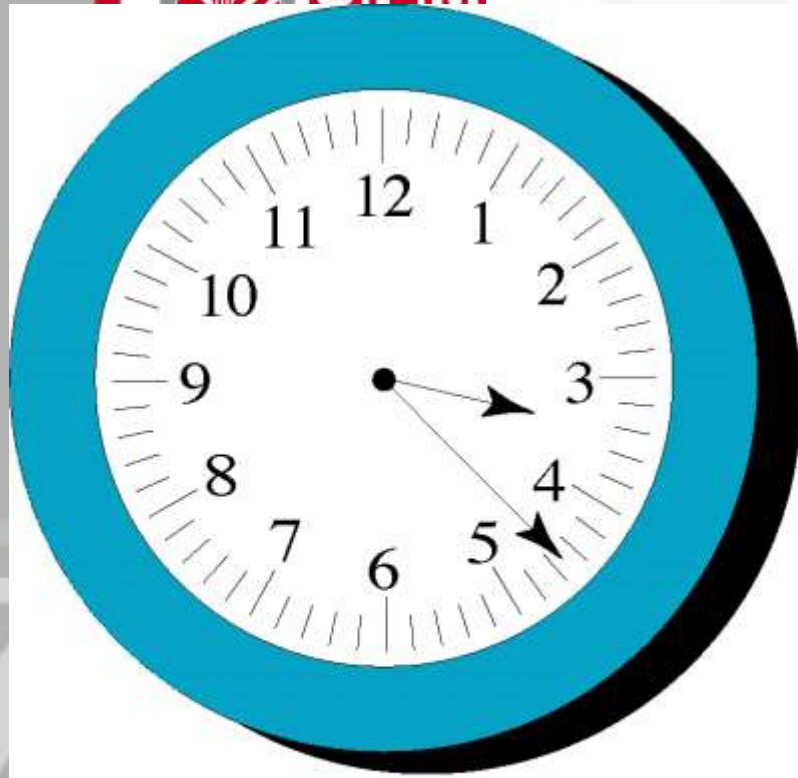
- Explain the data being sent in analog and digital signal (C4)
- Explain the circuit switching in telephone and cellular network(C4)
- Illustrates the unguided and guided transmission media(P2)



Analog and Digital Data

- Analog data – continuous information
 - I.e: analog clock – movement of hands are continuous
 - Human voice
- Digital data – discrete state information
 - I.e: digital clock – suddenly move from 8.05 – 8.06
 - Computer memory

Analog and Digital Clocks



a. Analog

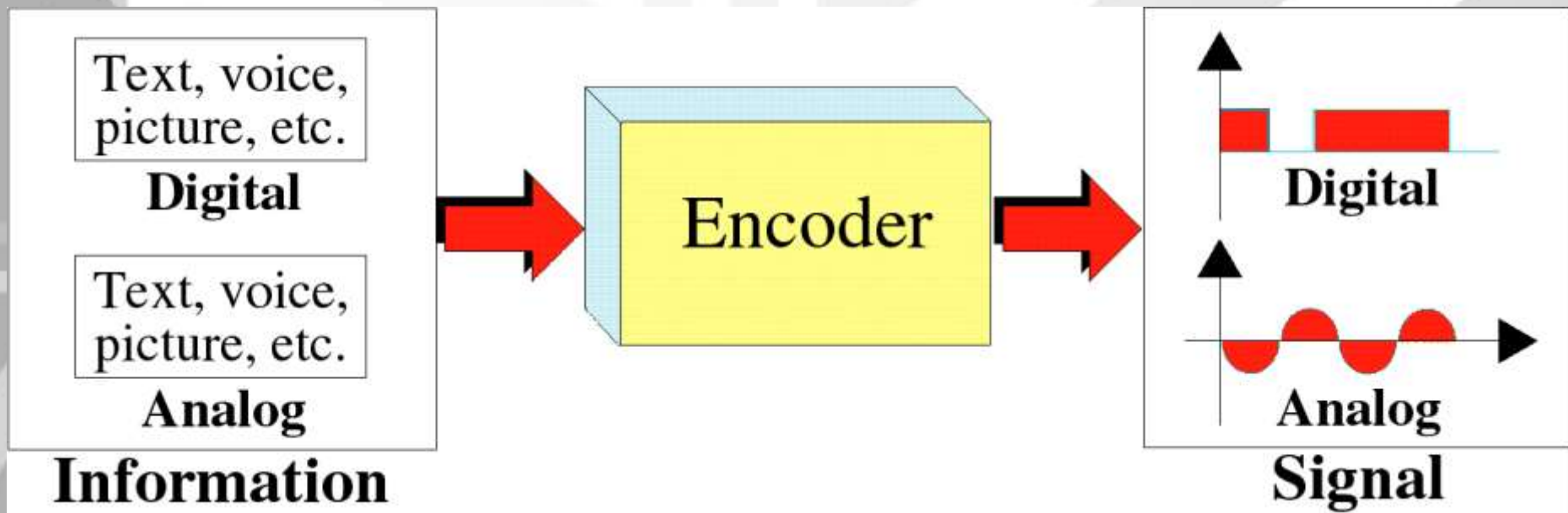


b. Digital

Figure 4-1

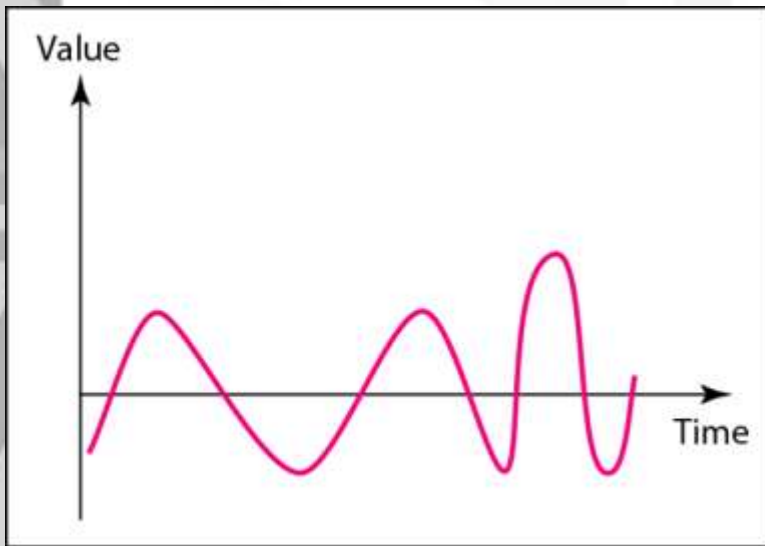


Transformation of Information to Signals

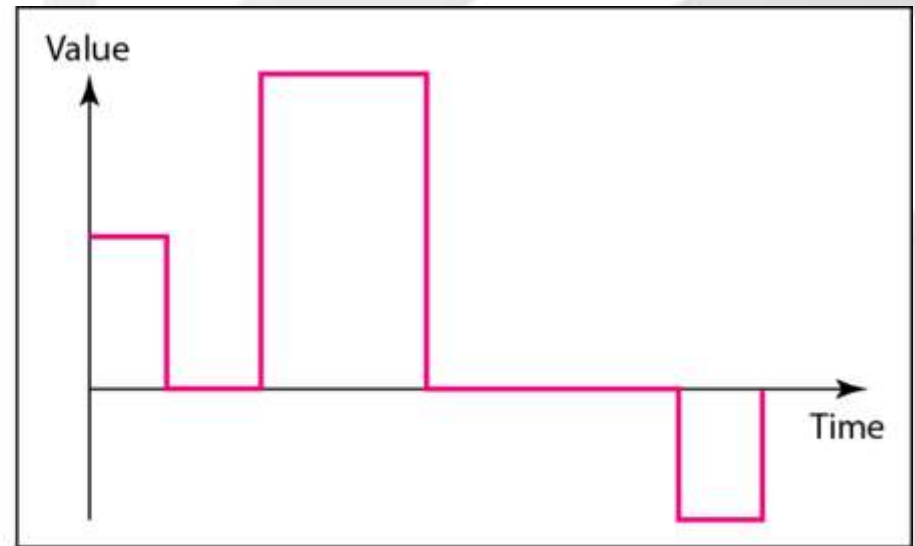


Analog and Digital Signal

- Analog signal – infinite number of values
- Digital signal – limited number of values



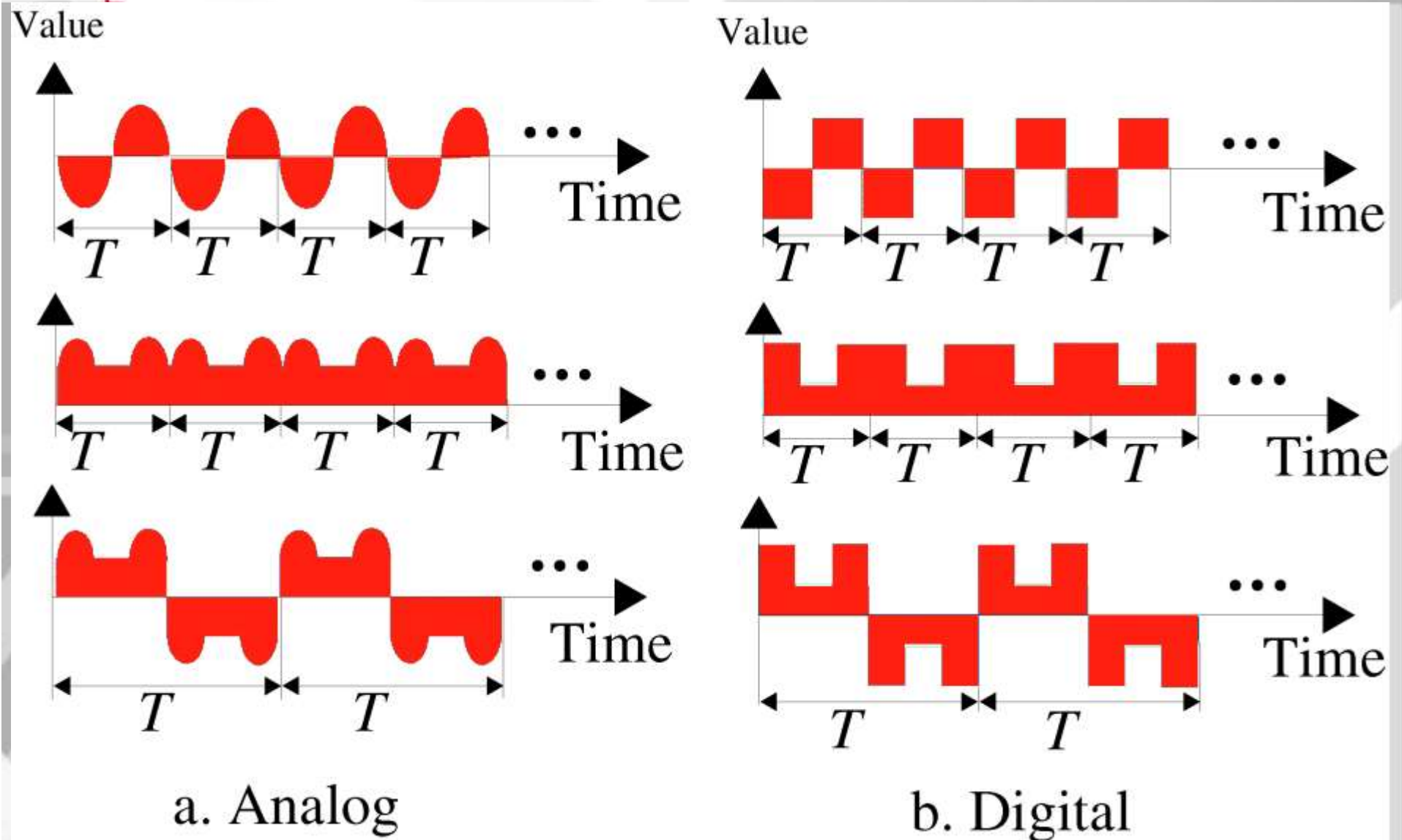
a. Analog signal



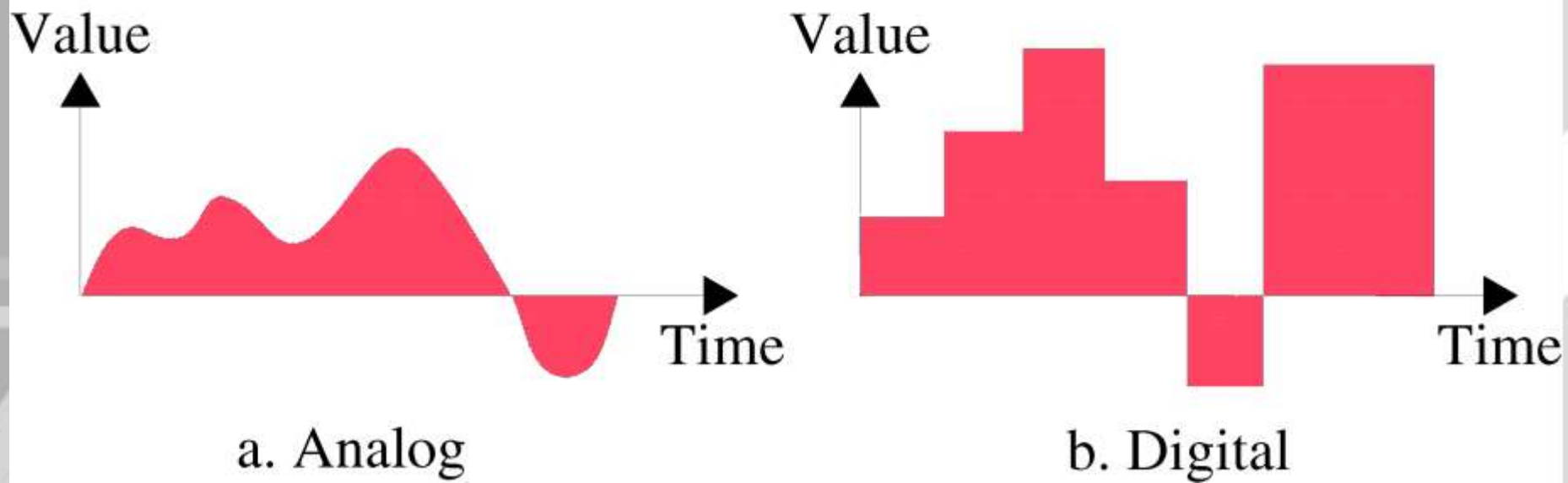
b. Digital signal

Figure 4-4

Periodic Signals



Aperiodic Signals





In data communications, we commonly use periodic analog signals and nonperiodic digital signals.

Periodic analog signals can be classified as **simple** or **composite**. A simple periodic analog signal, a **sine wave**, cannot be decomposed into simpler signals.

A composite periodic analog signal is composed of multiple sine waves.

Topic to be discussed:

- Sine Wave
- Phase
- Time and Frequency Domain
- Composite Signals
- Bandwidth

Figure 4-6

Sine Wave

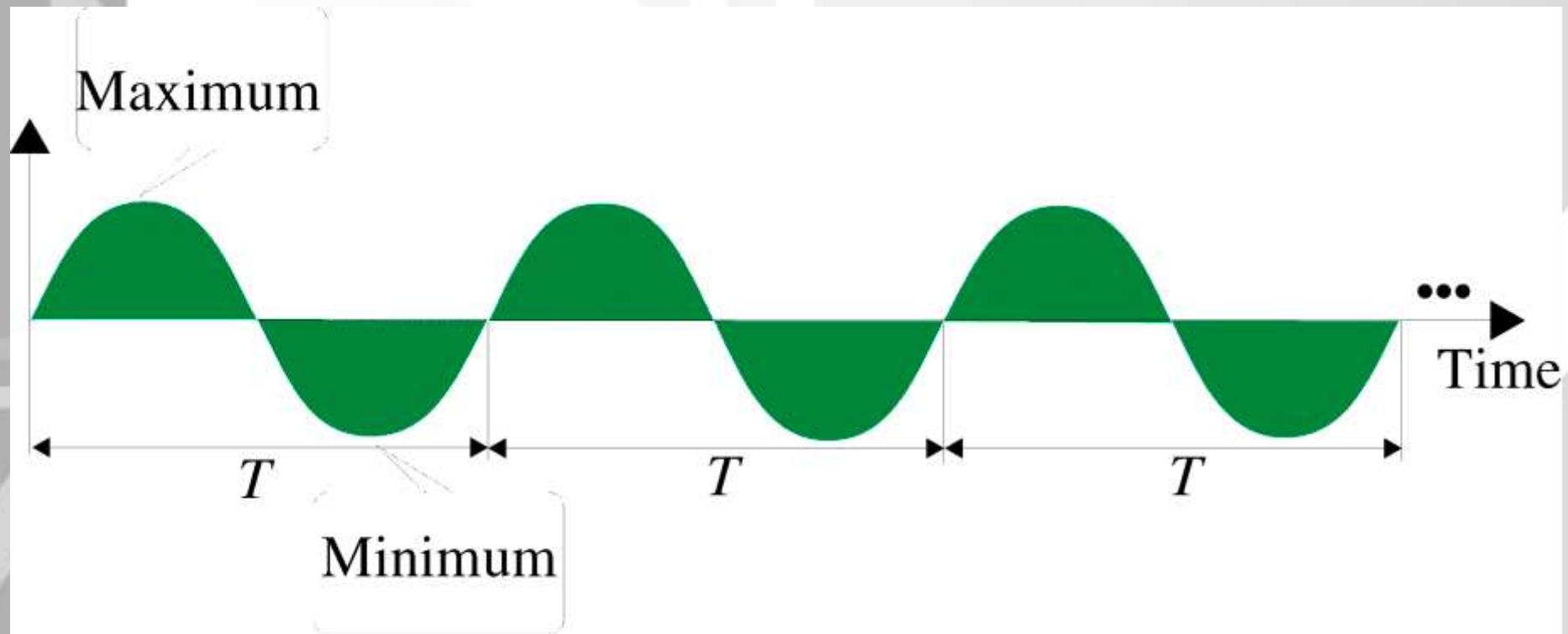


Figure 4-7

Phases

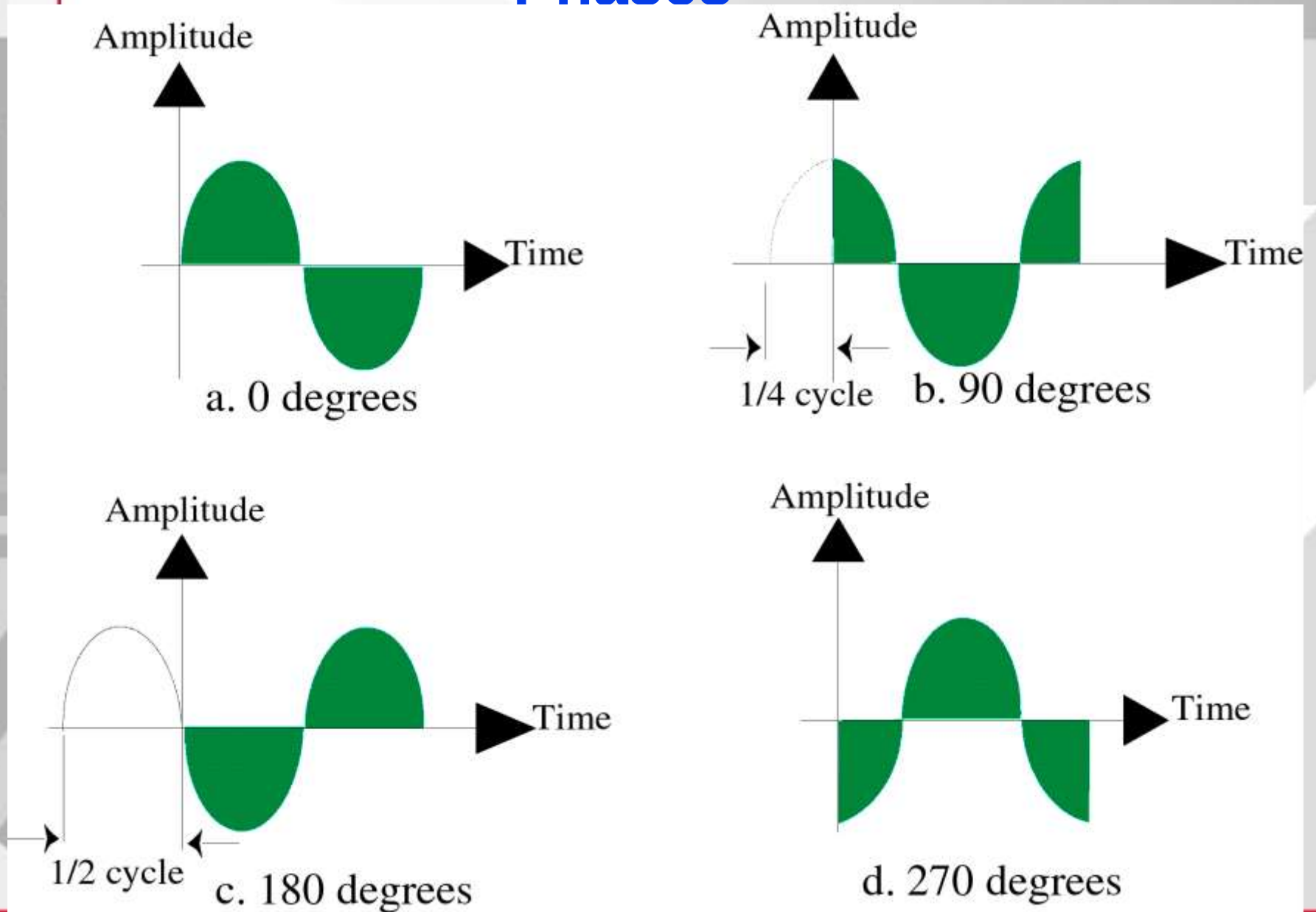
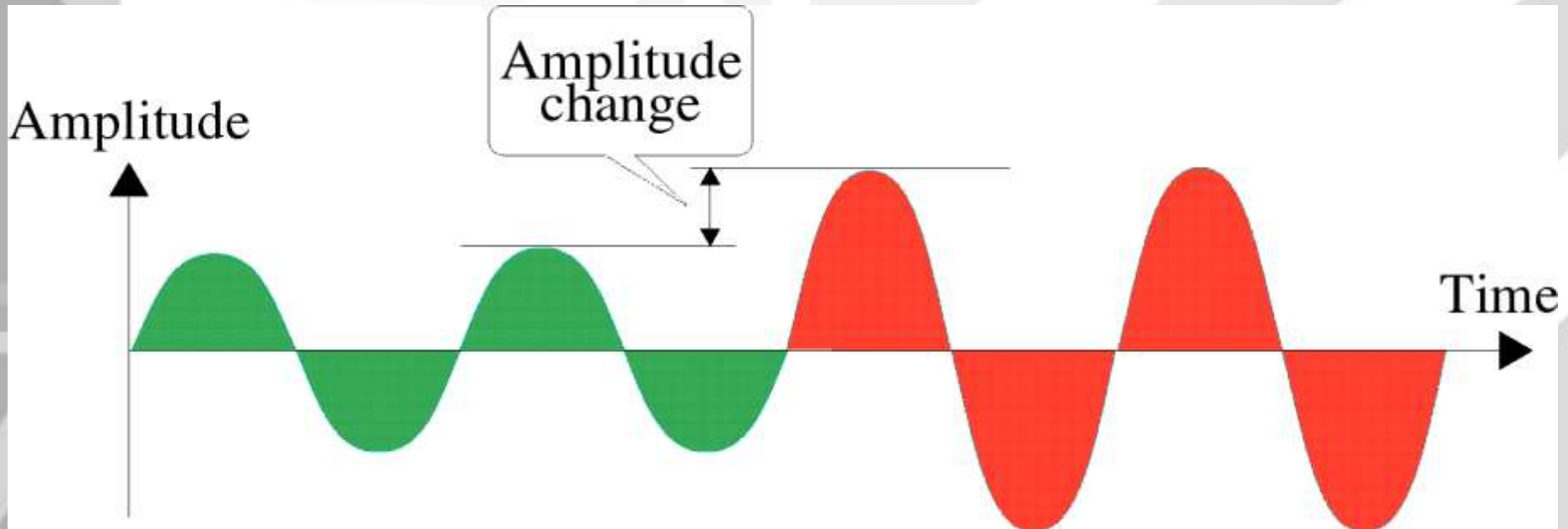


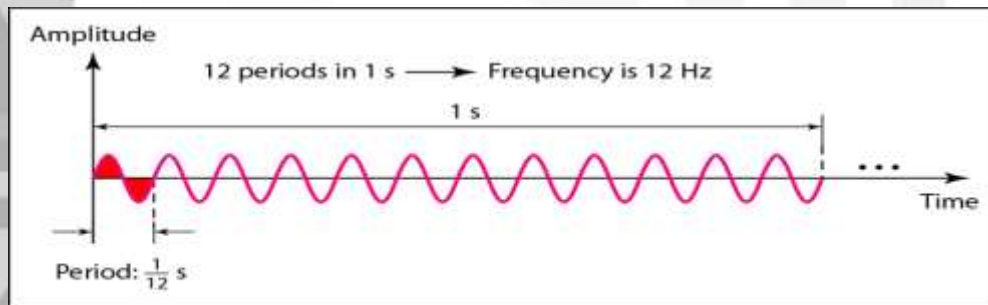
Figure 4-8

Amplitude Change

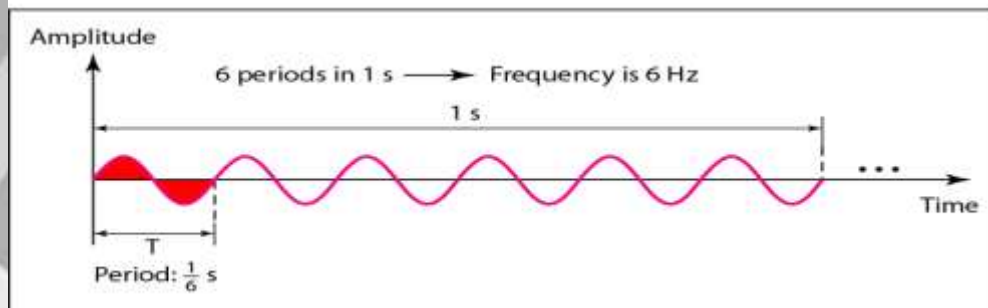


Period and Frequency

- Period – amount of time (in second) a signal need to complete 1 cycle
- Frequency – the number of cycles in 1 s



a. A signal with a frequency of 12 Hz



b. A signal with a frequency of 6 Hz

$$f = \frac{1}{T} \quad \text{and} \quad T = \frac{1}{f}$$

Period, $T = 1/12$ s

Period, $T = 1/6$ s

Example

- The power we use at home has a frequency of 60 Hz.
- The period of this sine wave?

$$f = \frac{1}{T} \quad \text{and} \quad T = \frac{1}{f}$$

$$T = \frac{1}{f} = \frac{1}{60} = 0.0166 \text{ s} = 0.0166 \times 10^3 \text{ ms} = 16.6 \text{ ms}$$

Units of period & frequency

Table 3.1 *Units of period and frequency*

<i>Unit</i>	<i>Equivalent</i>	<i>Unit</i>	<i>Equivalent</i>
Seconds (s)	1 s	Hertz (Hz)	1 Hz
Milliseconds (ms)	10^{-3} s	Kilohertz (kHz)	10^3 Hz
Microseconds (μ s)	10^{-6} s	Megahertz (MHz)	10^6 Hz
Nanoseconds (ns)	10^{-9} s	Gigahertz (GHz)	10^9 Hz
Picoseconds (ps)	10^{-12} s	Terahertz (THz)	10^{12} Hz

Figure 4-9



Frequency Change

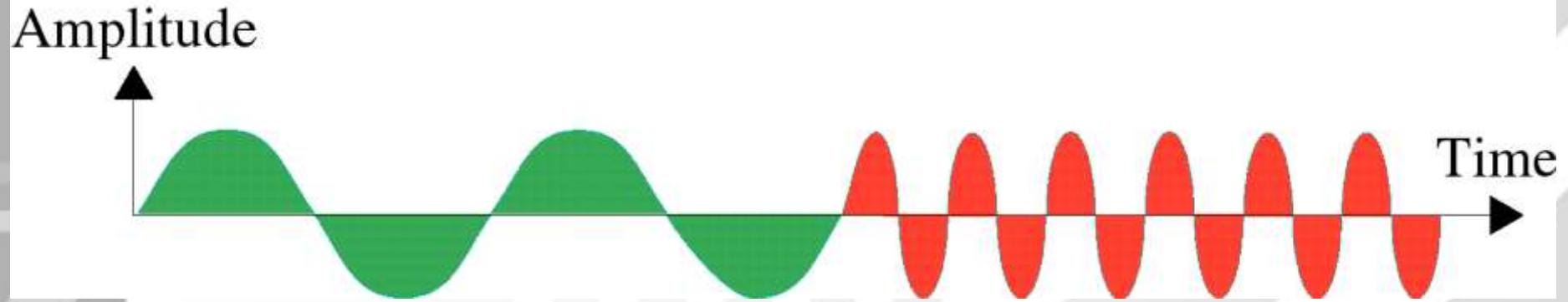
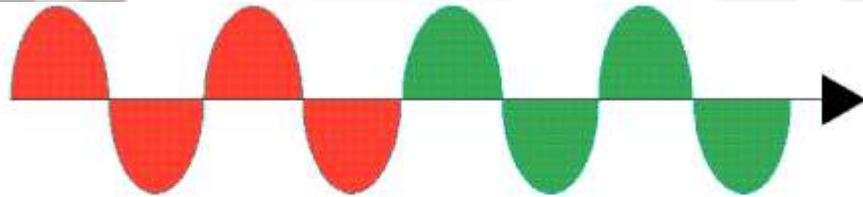
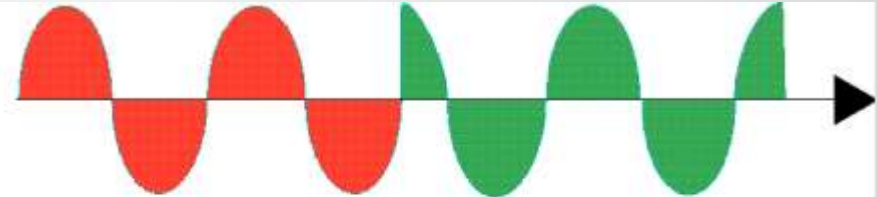


Figure 4-10

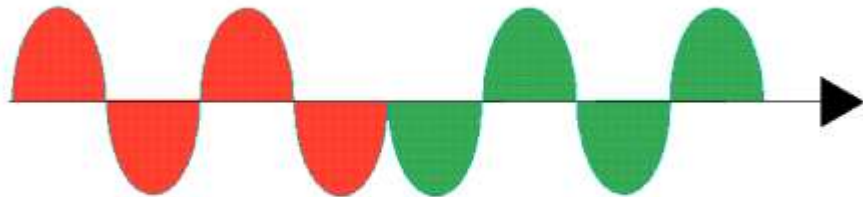
Phase Change



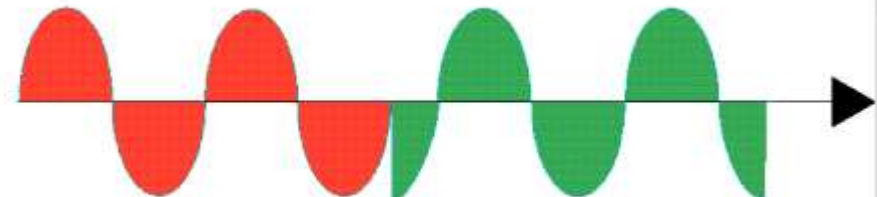
a. No phase change



b. 90 degree phase change



c. 180 degree phase change

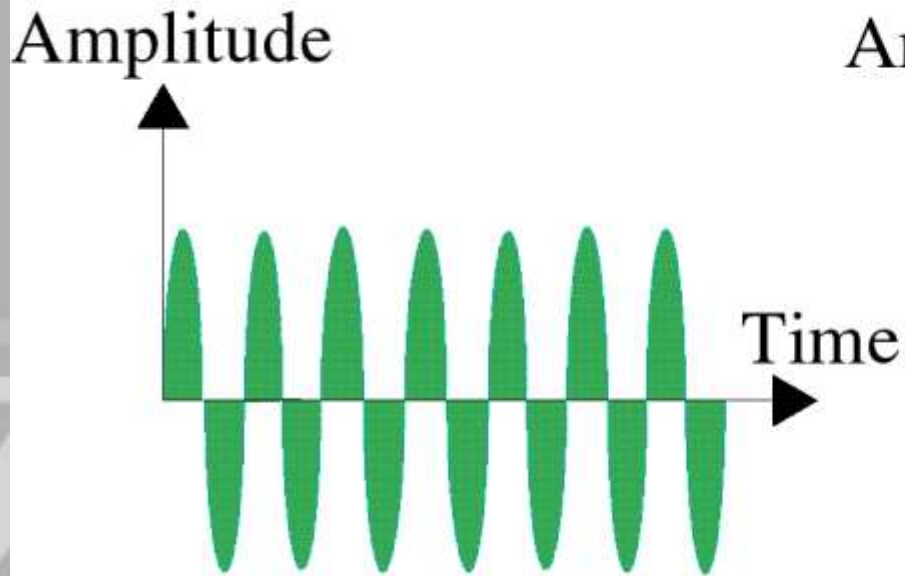


d. 270 degree phase change

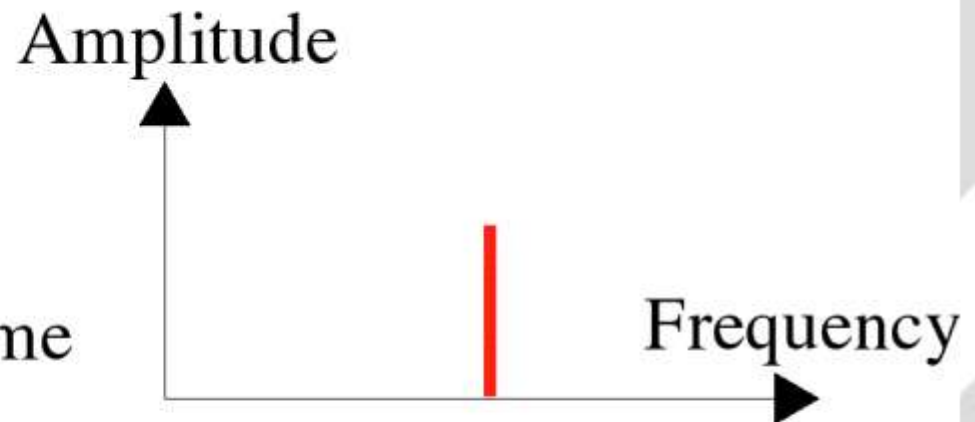
Figure 4-11



Time and Frequency Domain



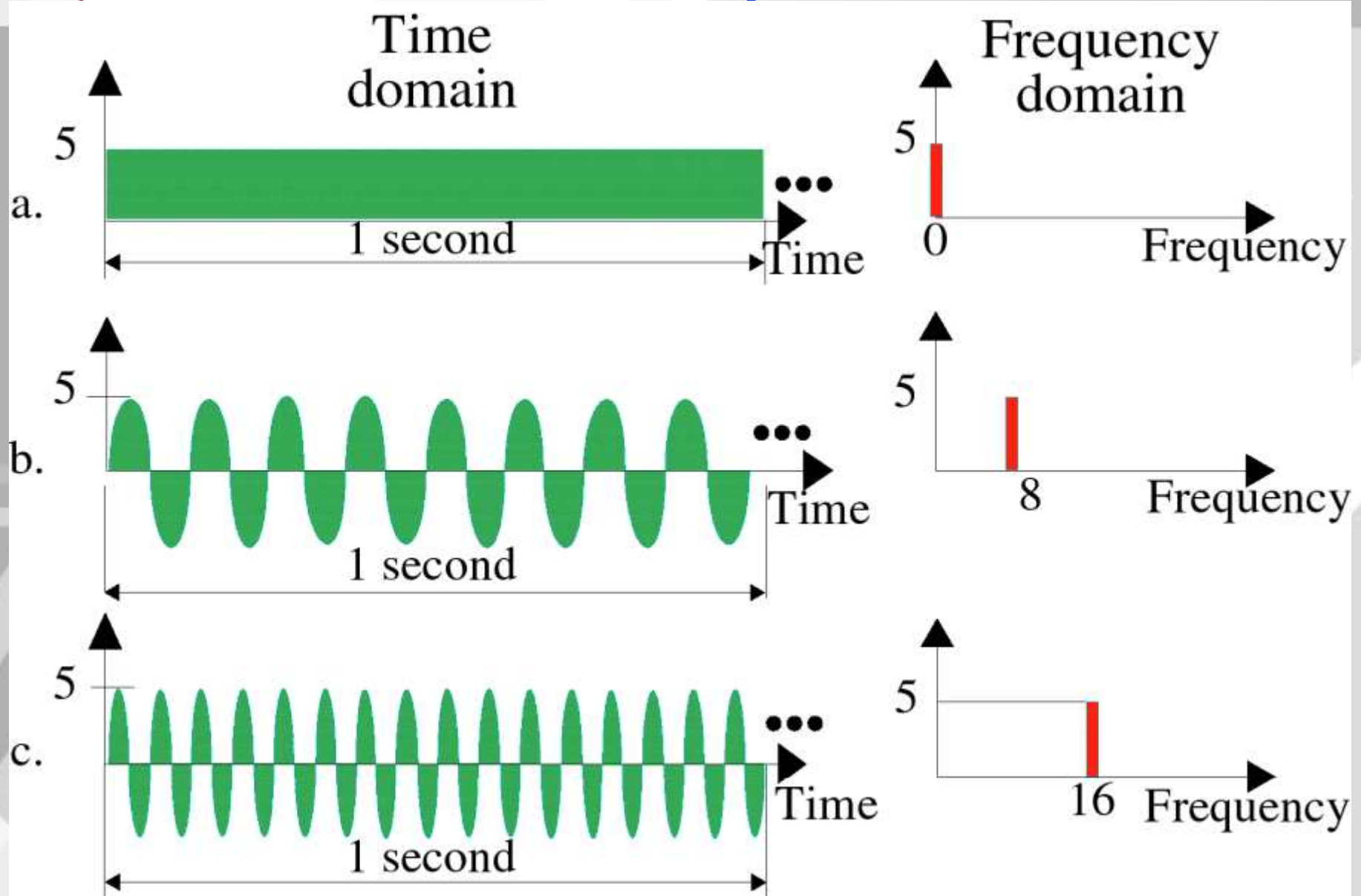
a. Time domain



b. Frequency domain

Figure 4-12

Examples



Frequency

- Frequency is the rate of change with respect to time.
- Change in a short span of time means high frequency.
- Change over a long span of time means low frequency.

Example 3.1



The power we use at home has a frequency of 60 Hz. The period of this sine wave can be determined as follows:

$$T = \frac{1}{f} = \frac{1}{60} = 0.0166 \text{ s} = 0.0166 \times 10^3 \text{ ms} = 16.6 \text{ ms}$$

Example 3.1



Express a period of 100 ms in microseconds.

Solution

The equivalents of 1 ms (1ms is 10^{-3} s) and 1s (1s is $10^6\mu\text{s}$)

$$\begin{aligned} 100\text{ms} &= 100 \times 10^{-3} \text{ s} \times 10^6\mu\text{s} \\ &= 10^2 \times 10^{-3} \text{ s} \times 10^6\mu\text{s} \\ &= 10^5\mu\text{s} \end{aligned}$$

Example 3.2



The period of a signal is 100 ms. What is its frequency in kilohertz?

Solution

First we change 100 ms to seconds, and then we calculate the frequency from the period ($1 \text{ Hz} = 10^{-3} \text{ kHz}$).

$$100 \text{ ms} = 100 \times 10^{-3} \text{ s} = 10^{-1} \text{ s}$$
$$f = \frac{1}{T} = \frac{1}{10^{-1}} \text{ Hz} = 10 \text{ Hz} = 10 \times 10^{-3} \text{ kHz} = 10^{-2} \text{ kHz}$$

Example

The voltage of a battery is a constant; this constant value can be considered a sine wave, as we will see later. For example, the peak value of an AA battery is normally 1.5 V.

Note

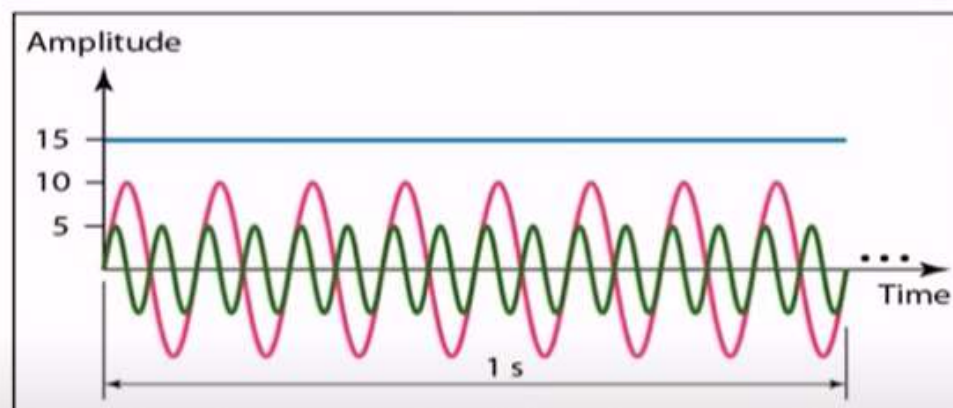
Frequency and period are the inverse each other.

$$f = \frac{1}{T} \quad \text{and} \quad T = \frac{1}{f}$$

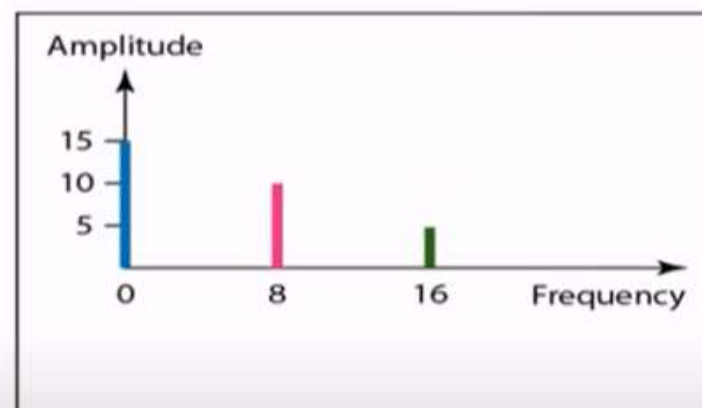
Note

If a signal does not change at all, its frequency is zero.
If a signal changes instantaneously, its frequency is infinite.

Figure 3.8 *The time domain and frequency domain of three sine waves*



a. Time-domain representation of three sine waves with frequencies 0, 8, and 16



b. Frequency-domain representation of the same three signals

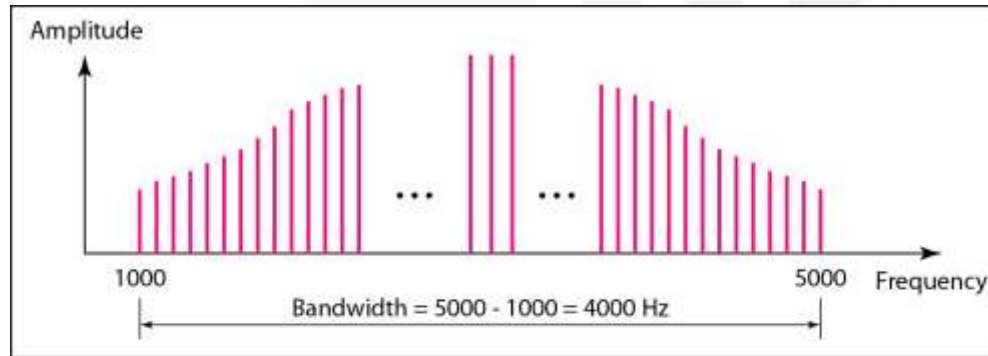
The frequency domain is more compact and useful when we are dealing with more than one sine wave. For example, Figure 3.8 shows three sine waves, each with different amplitude and frequency. All can be represented by three spikes in the frequency domain.

Note

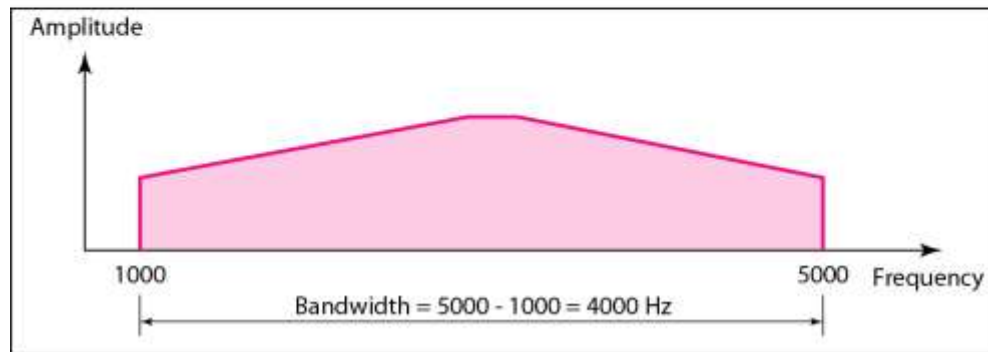
According to Fourier analysis, any composite signal is a combination of simple sine waves with different frequencies, amplitudes and phases.
Fourier analysis is discussed in Appendix C.

Bandwidth

- The bandwidth of a composite signal is the difference between the highest and the lowest frequencies contained in that signal.



a. Bandwidth of a periodic signal



b. Bandwidth of a nonperiodic signal

See example 3.10
and 3.11 in the
text book

Signals and Communication

- A single-frequency sine wave is not useful in data communications
- We need to send a composite signal, a signal made of many simple sine waves.
- According to Fourier analysis, any composite signal is a combination of simple sine waves with different frequencies, amplitudes, and phases.

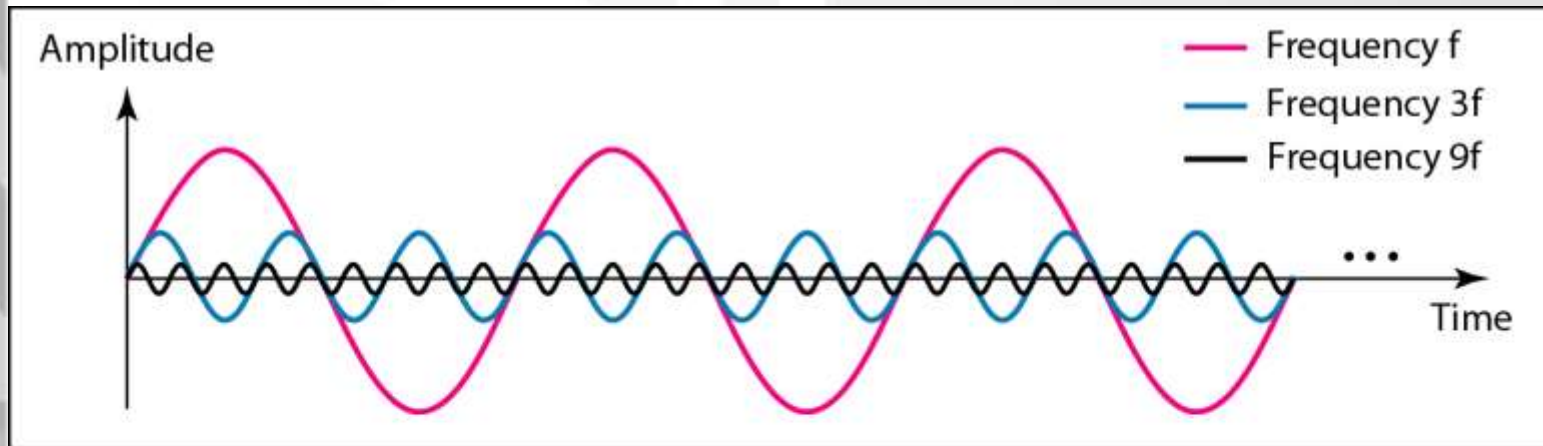


Composite Signals and Periodicity

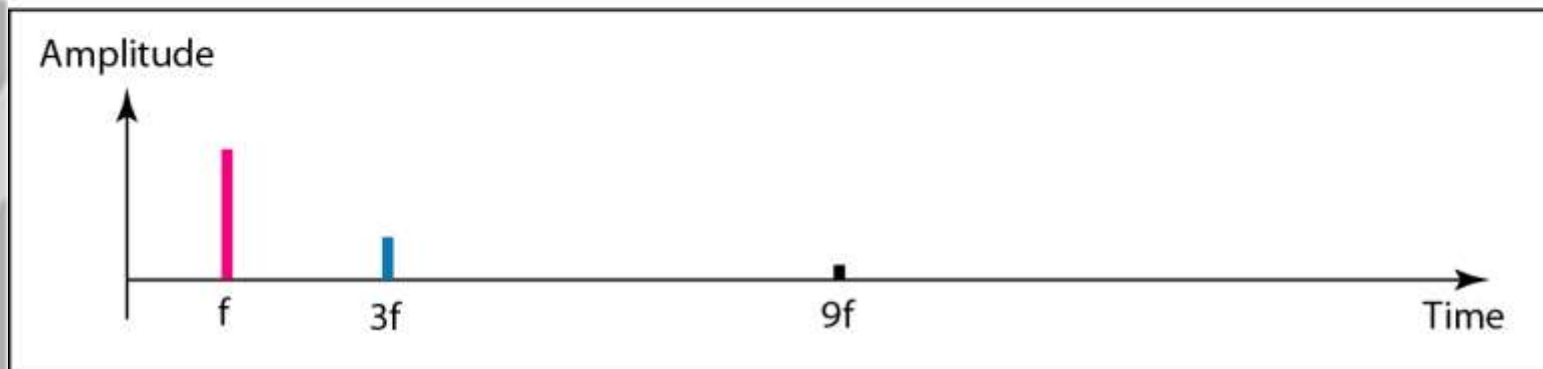
- If the composite signal is **periodic**, the decomposition gives a series of signals with **discrete** frequencies.
- If the composite signal is **nonperiodic**, the decomposition gives a combination of sine waves with **continuous** frequencies.



Figure 3.10 Decomposition of a composite periodic signal in the time and frequency domains

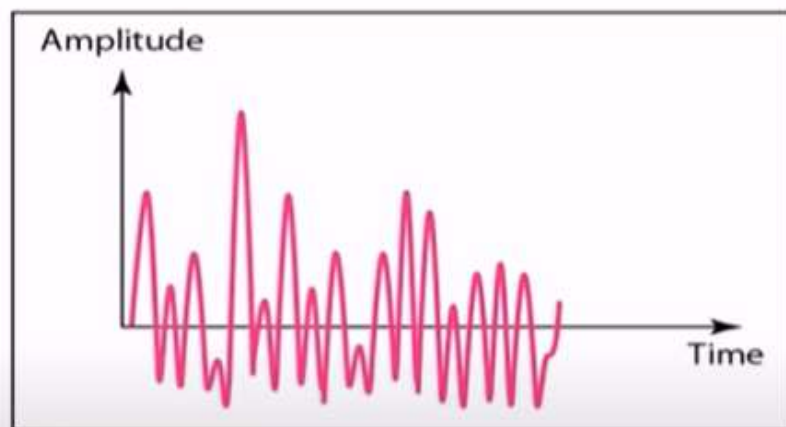


a. Time-domain decomposition of a composite signal

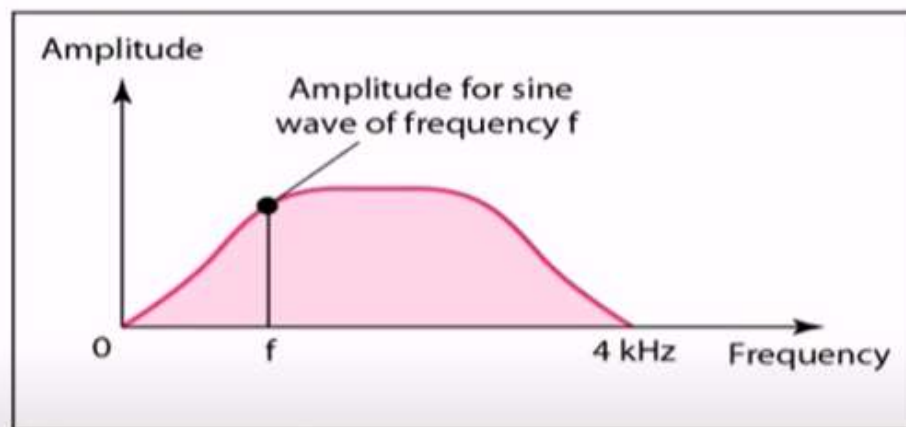


b. Frequency-domain decomposition of the composite signal

Figure 3.11 *The time and frequency domains of a nonperiodic signal*



a. Time domain



b. Frequency domain

Note

If the composite signal is periodic, the decomposition gives a series of signal discrete frequencies;

If the composite signal is nonperiodic, the decomposition gives a combination of sine waves with continuous frequencies.

Exercise

Example 3.10

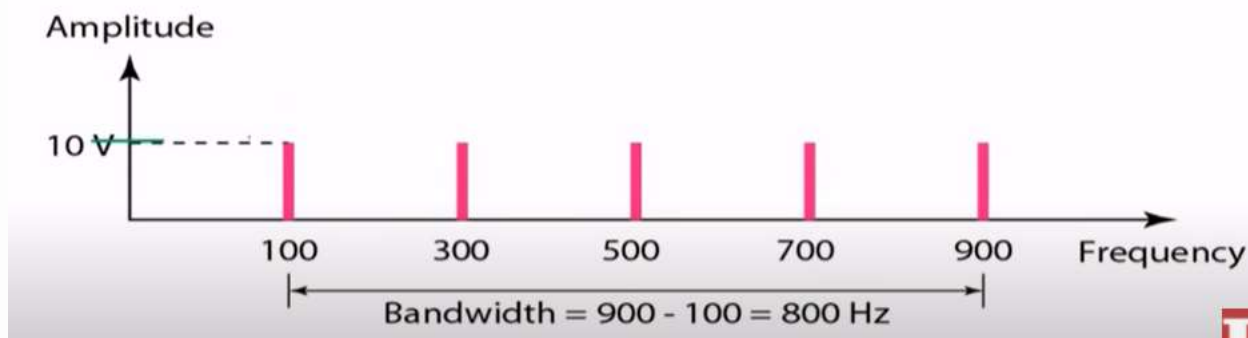
If a periodic signal is decomposed into five sine waves with frequencies of 100, 300, 500, 700, and 900 Hz, what is its bandwidth? Draw the spectrum, assuming all components have a maximum amplitude of 10 V.

Solution

Solution

Let f_h be the highest frequency, f_l the lowest frequency, and B the bandwidth. Then

$$B = f_h - f_l = 900 - 100 = 800 \text{ Hz}$$



Exercise

Example 3.11

A periodic signal has a bandwidth of 20 Hz. The highest frequency is 60 Hz. What is the lowest frequency? Draw the spectrum if the signal contains all frequencies of the same amplitude.

Solution

If the f_h be the highest frequency, f_l the lowest frequency and B is the bandwidth

$$B = f_h - f_l \Rightarrow 20 = 60 - f_l \Rightarrow f_l = 60 - 20 = 40 \text{ Hz}$$



Exercise

Example 3.11

A periodic signal has a bandwidth of 20 Hz. The highest frequency is 60 Hz. What is the lowest frequency? Draw the spectrum if the signal contains all frequencies of the same amplitude.

DIGITAL SIGNALS

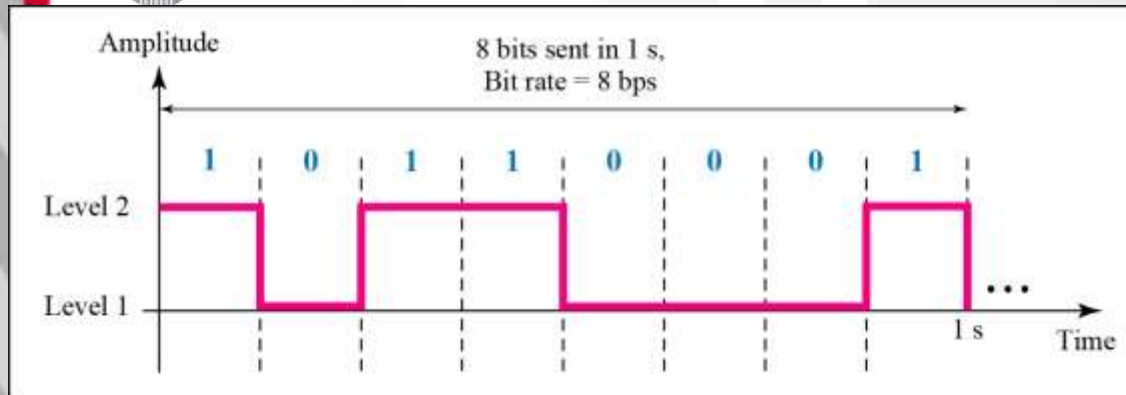
In addition to being represented by an analog signal, information can also be represented by a **digital signal**. For example, a 1 can be encoded as a positive voltage and a 0 as zero voltage. A digital signal can have more than two levels. In this case, we can send more than 1 bit for each level.

Topics discussed in this section:

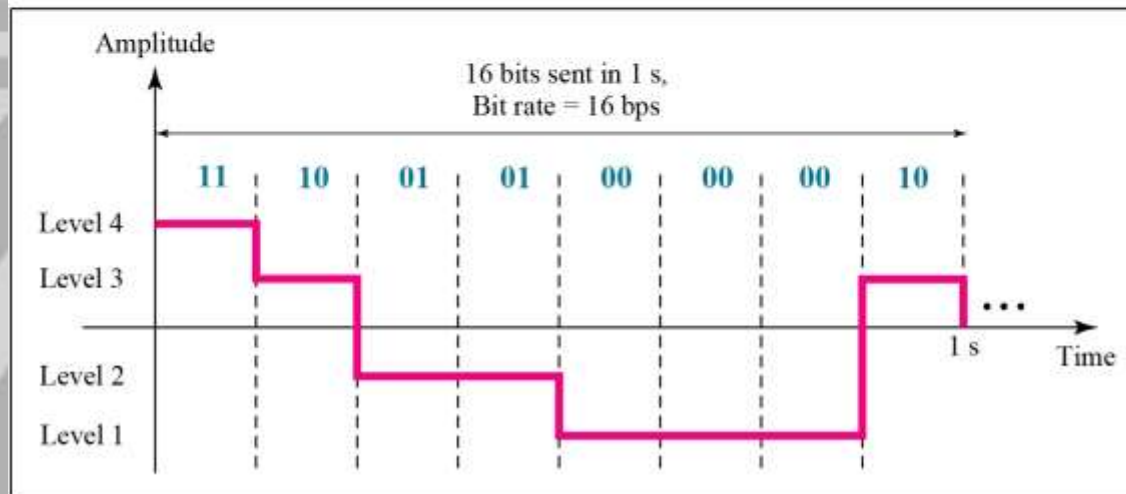
- Bit Rate

Digital Signal

- Bit rate :Number of bits in 1s (bit per second, bps)



a. A digital signal with two levels



b. A digital signal with four levels

Multilevel signalling
Use formula $\log_2 L$ bits to check for how many level required to send certain bits

Let level =8

How many bits per level?

Example 3.16



A digital signal has eight levels. How many bits are needed per level? We calculate the number of bits from the formula

$$\text{Number of bits per level} = \log_2 8 = 3$$

Each signal level is represented by 3 bits.

Example 3.17



A digital signal has nine levels. How many bits are needed per level? We calculate the number of bits by using the formula. Each signal level is represented by **3.17 bits**. However, this answer is **not realistic**. The number of bits sent per level needs to be an integer as well as a power of 2. For this example, 4 bits can represent one level.

Example 3.18



1. Assume we need to download text documents at the rate of 100 pages per **sec**. What is the required bit rate of the channel?

Hint:

A page is an average of 24 lines with 80 characters in each line. If we assume that one character requires 8 bits (ascii), the bit rate is

$$100 \times 24 \times 80 \times 8 = 1,536,000 \text{ bps} = 1.54 \text{ Mbps}$$

Example 3.18



2. Assume we need to download text documents at the rate of 200 pages per min. What is the required bit rate of the channel?
3. Assume your ISP downloading rate is 55Mbps. How many pages your device able to download in a minute?
4. Assume your ISP downloading rate is only 10Mbps. How many pages your device able to download in a second if the encoding system is is UNICODE?

03 DATA COMMUNICATIONS AND
NETWORKING Data and Signals Part 2 -
YouTube

Example 3.18 (Answer)



2. Assume we need to download text documents at the rate of 200 pages per min. What is the required bit rate of the channel?

$$\begin{aligned} 200 \text{ pages}/60\text{s} &= 3.333 \text{ pages/second} \\ &= 3.333 \text{ pages} \times 24 \text{ lines} \times 80 \text{ characters} \times 8 \text{ bits} = 51194.88 \text{ bps} \\ &= 51194.88\text{bps} \times 10^{-6} \text{ Mbps} \\ &= 0.05119 \text{ Mbps} \end{aligned}$$

3. Assume your ISP downloading rate is 55Mbps. How many pages your device able to download in a minute?

$$\begin{aligned} 55\text{Mbps} \\ &= 55\,000\,000 / (24 \text{ lines} \times 80 \text{ characters} \times 8 \text{ bits}) \\ &= 3580.73 \text{ pages} \times 60 \text{ s} \\ &= 214844 \text{ page per minutes} \end{aligned}$$

Example 3.18 (Answer)



4. Assume your ISP downloading rate is only 10Mbps. How many pages your device able to download in a second if the encoding system is UNICODE? (Unicode is 16 bits)

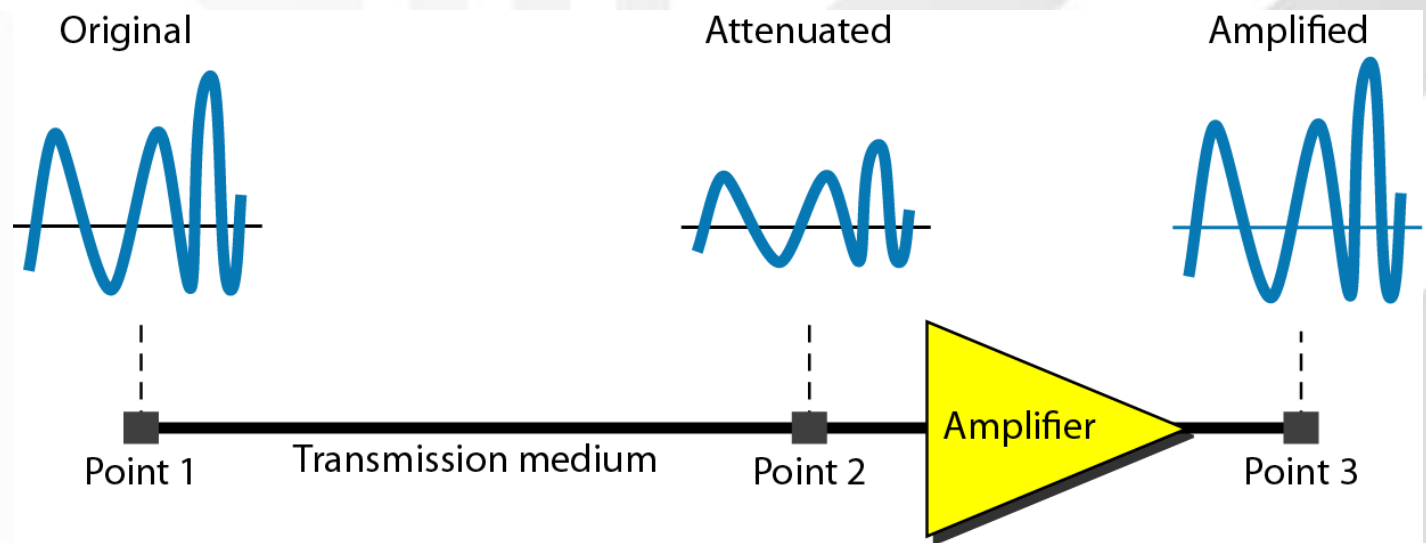


Transmission Impairment

- Signals travel through transmission media, which are not perfect.
- The imperfection causes signal impairment.
- This means that the signal at the beginning of the medium is not the same as the signal at the end of the medium. What is sent is not what is received.
- Three causes of impairment are attenuation, distortion, and noise

Attenuation

- Loss of energy- due to resistance of the medium
- Energy, converted to heat



- Measured in decibel (dB) = $10\log_{10} (P_2/P_1)$
- See example 3.26 – 3.30 for self study

Example 3.26



Suppose a signal travels through a transmission medium and its power is reduced to one-half. This means that P_2 is $(1/2)P_1$. In this case, the attenuation (loss of power) can be calculated as

$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{0.5P_1}{P_1} = 10 \log_{10} 0.5 = 10(-0.3) = -3 \text{ dB}$$

A loss of 3 dB (–3 dB) is equivalent to losing one-half the power.

Example 3.27



A signal travels through an amplifier, and its power is increased 10 times. This means that $P_2 = 10P_1$. In this case, the amplification (gain of power) can be calculated as

$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{10P_1}{P_1}$$

$$= 10 \log_{10} 10 = 10(1) = 10 \text{ dB}$$

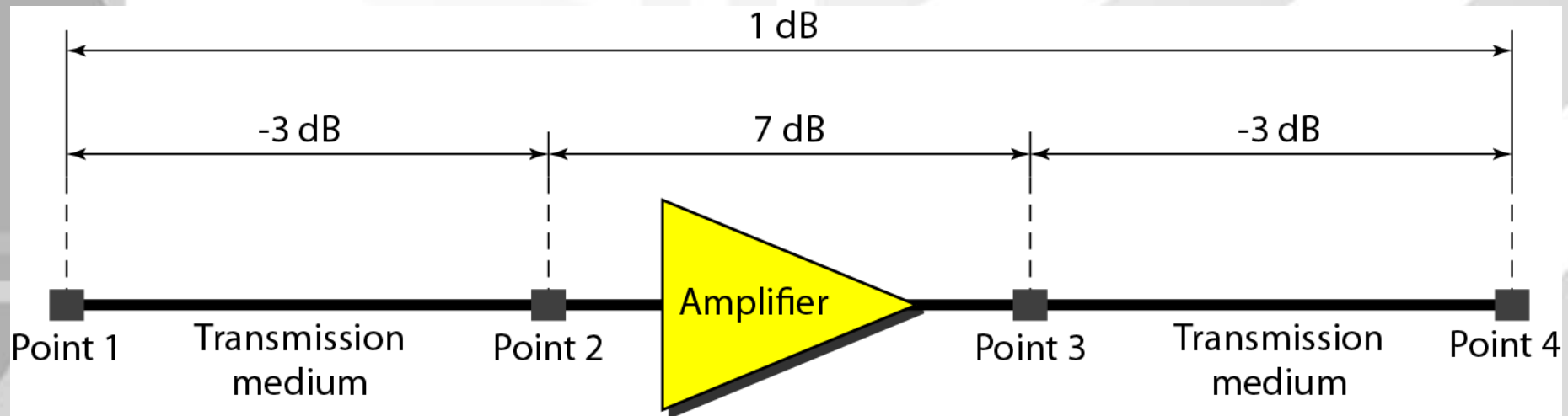
Example 3.28



One reason that engineers use the decibel to measure the changes in the strength of a signal is that decibel numbers can be added (or subtracted) when we are measuring several points (cascading) instead of just two. In Figure 3.27 a signal travels from point 1 to point 4. In this case, the decibel value can be calculated as

$$\text{dB} = -3 + 7 - 3 = +1$$

Figure 3.27 Decibels for Example 3.28



Example 3.29



Sometimes the decibel is used to measure signal power in milliwatts. In this case, it is referred to as dB_m and is calculated as $\text{dB}_m = 10 \log_{10} P_m$, where P_m is the power in milliwatts. Calculate the power of a signal with $\text{dB}_m = -30$.

Solution

We can calculate the power in the signal as

$$\begin{aligned}\text{dB}_m &= 10 \log_{10} P_m = -30 \\ \log_{10} P_m &= -3 \quad P_m = 10^{-3} \text{ mW}\end{aligned}$$

Example 3.30



The loss in a cable is usually defined in decibels per kilometer (dB/km). If the signal at the beginning of a cable with -0.3 dB/km has a power of 2 mW, what is the power of the signal at 5 km?

Solution

The loss in the cable in decibels is $5 \times (-0.3) = -1.5$ dB. We can calculate the power as

$$\text{dB} = 10 \log_{10} \frac{P_2}{P_1} = -1.5$$

$$\frac{P_2}{P_1} = 10^{-0.15} = 0.71$$

$$P_2 = 0.71 P_1 = 0.7 \times 2 = 1.4 \text{ mW}$$



EXERCISES:

1 If the signal at the end of a cable with -0.2 dB/km has a power of 100 mW for 10km, what is the power of the signal at :

1. 3 km
2. 0 km
3. $1 \text{ km} \log_{10} P_2/P_1$



2. A signal that has been transmitted to the user has experienced attenuation in the network causing loss of signal strength. If the signal at the beginning of a cable has a power of 25W and signal at the receiving end has a power of 500mW with a distance of 30km between each other, calculate:

a. loss of power for every kilometer.

b. power of signal at first 1km.

Note: Show your calculations.

$$\text{Signal Loss in dB} = 10 \log_{10} (P_2/P_1)$$

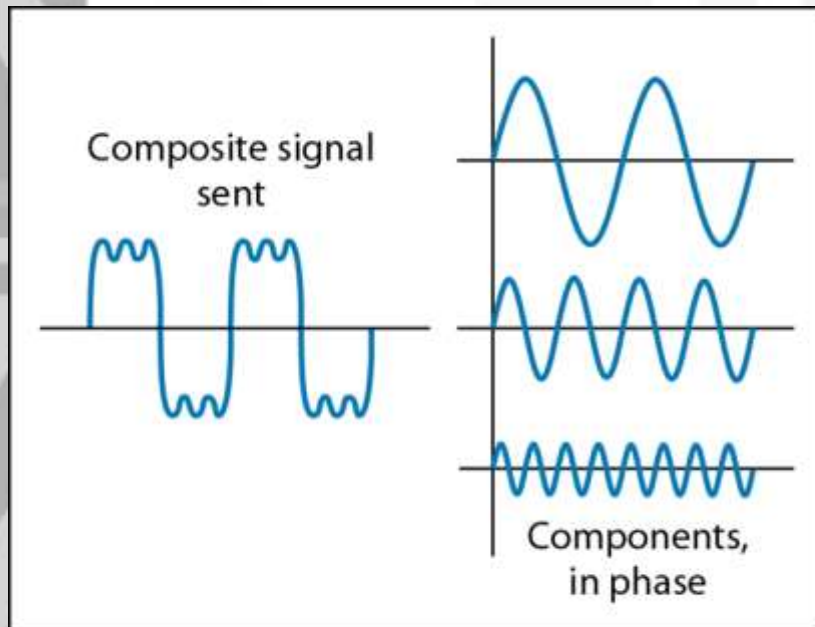
3. Attenuation occurs on networks caused a loss of signal strength. If the signal at the beginning of a cable has a power of 50W and signal at the receiving end has a power of 200mW with a loss of power for every km is -0.5 dB , calculate:

a. How long the distance between beginning of node and receiving node.

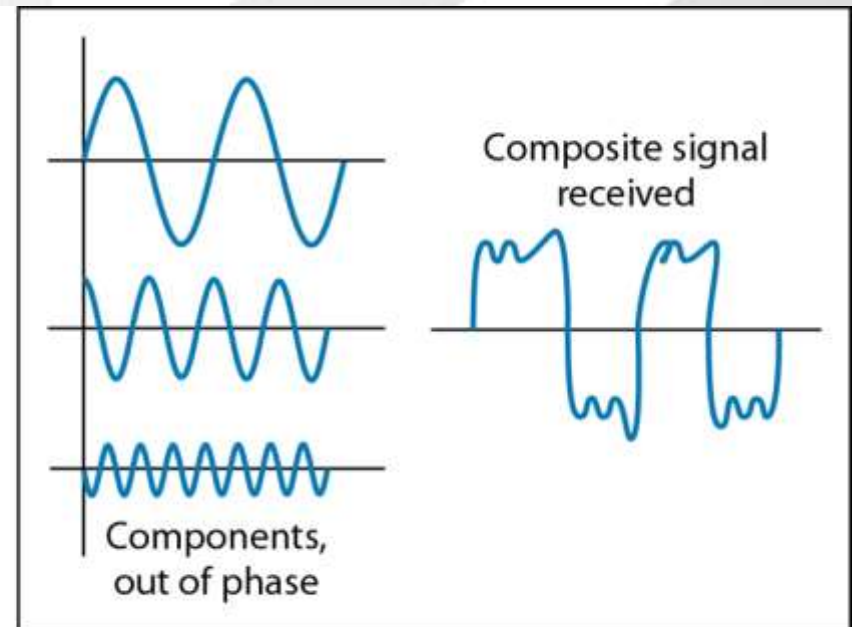
b. power of signal at first 3km

Distortion

- Signal changes its form or shape
- Due to different frequency of composite signal



At the sender



At the receiver

Noise

- There are different types of noise
 - **Thermal** - random noise of electrons in the wire creates an extra signal
 - **Induced** - from motors and appliances, devices act as transmitter antenna and medium as receiving antenna.
 - **Crosstalk** - same as above but between two wires.
 - **Impulse** - Spikes that result from power lines, lightning, etc.

